MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, ARITHMETIC LOGIC UNITS/FUNCTION GENERATORS MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, arithmetic logic units/function generators. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Arithmetic logic unit/function generator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline, (see MIL-M-38510, appendix C)
J	D-3 (24-lead, 1/2" x 1-1/4", dual-in-line package)
K	F-6 (24-lead, 3/8" x 5/8", flat package)
L	F-7 (24-lead, 3/8" x 1/2", flat package)
Z	F-8 (24-lead, 1/4" x 3/8", flat package)

1.3 Absolute maximum ratings.

Supply voltage	-0.5 to 7.0 Vdc
Input voltage range	-1.5 Vdc at -18 mA to 5.5 Vdc
Storage temperature range	-65° to 150°C
Maximum power dissipation, P_D 1/	193 mWdc
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC})	0.04°C/mW for flat package; 0.02°C/mW for dual-in-line package
Junction temperature (T_J)	175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage	4.5 minimum to 5.5 Vdc maximum
Minimum high-level input voltage	2.0 Vdc
Maximum low-level input voltage	0.7 Vdc
Normalized fanout (each output)	
Logical low level	10 maximum
Logical high level	20 maximum
Case operating temperature range	-55° to 125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Case outline. Case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.2.2 Terminal connections and pin designations. The terminal connections and pin designations shall be as specified on figure 1.

3.2.3 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.4 Logic diagram. The logic diagram shall be as specified on figure 3.

3.2.5 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Condition 1/	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; V_{IL} = 0.7 V; V_{IH} = 2.0 V; I_{OH} = -400 µA	01	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IL} = 0.7 V; V_{IH} = 2.0 V; I_{OL} = 4 mA	01		0.4	V
Input diode clamp	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -18 mA T_C = 25°C	01		-1.5	V
Low-level input current at M input	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 2/	01	-0.03	-0.40	mA
Low-level input current at $\overline{B}$ input	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 2/	01	-0.09	-1.20	mA
Low-level input current at S_0 or S_3 input	I_{IL3}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 2/	01	-0.12	-1.60	mA
Low-level input current at S_1 or S_2	I_{IL4}	V_{CC} = 5.5 V; V_{IN} = 0.4 V A_0, A_1, A_2, A_3 = 5.5 V 3/	01	-0.12	-1.60	mA
Low-level input current at C_n input	I_{IL5}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 3/	01	-0.15	-2.0	mA
Low-level input current at $\overline{A}$ input	I_{IL6}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 2/	01	-0.09	-1.20	mA
High-level input current at M input	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 3/	01		20	µA
High-level input current at $\overline{A}$ or $\overline{B}$ input	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 3/	01		60	µA
High-level input current at S_0 or S_3 input	I_{IH3}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 3/	01		80	µA
High-level input current at S_1 or S_2	I_{IH4}	V_{CC} = 5.5 V; V_{IN} = 2.7 V B_0, B_1, B_2, B_3 = 5.5 V 3/	01		80	µA
High-level input current at C_n input	I_{IH5}	V_{CC} = 5.5 V; V_{IN} = 2.7 V 2/	01		100	µA
High-level input current at M input	I_{IH6}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 3/	01		100	µA
High-level input current at $\overline{A}$ or $\overline{B}$ input	I_{IH7}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 3/	01		300	µA
High-level input current at S_0 or S_3 input	I_{IH8}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 3/	01		400	µA
High-level input current at S_1 or S_2	I_{IH9}	V_{CC} = 5.5 V; V_{IN} = 5.5 V; $\overline{B}_0, \overline{B}_1, \overline{B}_2, \overline{B}_3$ = 5.5 V 3/	01		400	µA
High-level input current at C_n input	I_{IH10}	V_{CC} = 5.5 V; V_{IN} = 5.5 V 2/	01		500	µA
Short-circuit output current at $\overline{G}$ output	I_{OS1}	V_{CC} = 5.5 V; $S_3, \overline{A}_3, \overline{B}_3$ = 5.5 V 3/4/	01	-15	-100	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Condition 1/	Device type	Limits Min	Limits Max	Unit
Short-circuit output current at C_{n+4} output	I_{OS2}	V_{CC} = 5.5 V; 3/ 4/	01	-15	-100	mA
Short-circuit output current at $\overline{P}$, $\overline{F}_0$ thru $\overline{F}_3$ outputs	I_{OS3}	V_{CC} = 5.5 V; 2/ 4/	01	-15	-100	mA
Collector cutoff current at A = B output	I_{CEX}	V_{CC} = 4.5 V; $V_{(A=B)}$ = 5.5 V; all input = 4.5 V	01		100	µA
Supply current	I_{CC1}	S_0 thru S_3, M and A inputs = 5.5 V; V_{CC} = 5.5 V	01		35	mA
	I_{CC2}	V_{CC} = 5.5 V; S_0 thru S_3 and M = 5.5 V; all others grounded	01		35	mA
Propagation delay times, high-to-low level output: 5/ 6/						
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{sum}$ mode	t_{PHL1}	V_{CC} = 5.0 V See figure 4 $\overline{sum}$ mode test table	01	2	37	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{diff}$ mode	t_{PHL2}	V_{CC} = 5.0 V See figure 4 $\overline{diff}$ mode test table	01	2	55	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{P}$ $\overline{sum}$ mode	t_{PHL3}	V_{CC} = 5.0 V See figure 4 $\overline{sum}$ mode test table	01	2	52	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{P}$ $\overline{diff}$ mode	t_{PHL4}	V_{CC} = 5.0 V See figure 4 $\overline{diff}$ mode test table	01	2	57	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{sum}$ mode	t_{PHL5}	V_{CC} = 5.0 V See figure 4 $\overline{sum}$ mode test table	01	2	42	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{diff}$ mode	t_{PHL6}	V_{CC} = 5.0 V See figure 4 $\overline{diff}$ mode test table	01	2	55	ns
C_n to any $\overline{F}$ $\overline{sum}$ mode	t_{PHL7}	V_{CC} = 5.0 V See figure 4 $\overline{sum}$ mode test table	01	2	37	ns
$\overline{A}_i$ to $\overline{B}_i$ to A = B $\overline{diff}$ mode	t_{PHL8}	V_{CC} = 5.0 V See figure 4 $\overline{diff}$ mode test table	01	2	100	ns
C_n to C_{n+4} $\overline{sum}$ mode	t_{PHL9}	V_{CC} = 5.0 V See figure 4 $\overline{sum}$ mode test table	01	2	37	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Condition 1/	Device type	Limits Min	Limits Max	Unit
Propagation delay times, high-to-low level output: 5 6/						
C_n to C_{n+4} $\overline{\text{diff}}$ mode	t_{PHL10}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	37	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{C}_{n+4}$ $\overline{\text{sum}}$ mode	t_{PHL11}	V_{CC} = 5.0 V See figure 4 $\overline{\text{sum}}$ mode test table	01	2	64	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{C}_{n+4}$ $\overline{\text{diff}}$ mode	t_{PHL12}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	60	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ logic mode	t_{PHL13}	V_{CC} = 5.0 V See figure 4 logic mode test table	01	2	48	ns
Propagation delay times, low-to-high level output: 5/ 6/						
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{\text{sum}}$ mode	t_{PLH1}	V_{CC} = 5.0 V See figure 4 $\overline{\text{sum}}$ mode test table	01	2	55	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ $\overline{\text{diff}}$ mode	t_{PLH2}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	55	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{P}$ $\overline{\text{sum}}$ mode	t_{PLH3}	V_{CC} = 5.0 V See figure 4 $\overline{\text{sum}}$ mode test table	01	2	52	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{P}$ $\overline{\text{diff}}$ mode	t_{PLH4}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	52	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{\text{sum}}$ mode	t_{PLH5}	V_{CC} = 5.0 V See figure 4 $\overline{\text{sum}}$ mode test table	01	2	51	ns
$\overline{A}_i$ or $\overline{B}_i$ to $\overline{G}$ $\overline{\text{diff}}$ mode	t_{PLH6}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	55	ns
C_n to any $\overline{F}$ $\overline{\text{sum}}$ mode	t_{PLH7}	V_{CC} = 5.0 V See figure 4 $\overline{\text{sum}}$ mode test table	01	2	46	ns
$\overline{A}_i$ or $\overline{B}_i$ to A = B $\overline{\text{diff}}$ mode	t_{PLH8}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	76	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Condition	Device type	Limits Min	Limits Max	Unit
Propagation delay times, low-to-high level output: 5/ 6/						
C_n to C_{n+4} $\overline{\text{sum}}$ mode	t_{PLH9}	V_{CC} = 5.0 V See figure 4 $\overline{\text{sum}}$ mode test table	01	2	48	ns
C_n to C_{n+4} $\overline{\text{diff}}$ mode	t_{PLH10}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	48	ns
$\overline{A}_i$ or $\overline{B}_i$ to C_{n+4} $\overline{\text{sum}}$ mode	t_{PLH11}	V_{CC} = 5.0 V See figure 4 $\overline{\text{sum}}$ mode test table	01	2	64	ns
$\overline{A}_i$ or $\overline{B}_i$ to C_{n+4} $\overline{\text{diff}}$ mode	t_{PLH12}	V_{CC} = 5.0 V See figure 4 $\overline{\text{diff}}$ mode test table	01	2	69	ns
$\overline{A}_i$ or $\overline{B}_i$ to any $\overline{F}$ logic mode	t_{PLH13}	V_{CC} = 5.0 V See figure 4 logic mode test table	01	2	57	ns

1/ Condition of inputs specified in table III.

2/ All unspecified inputs at 5.5 V.

3/ All unspecified inputs at 0 V.

4/ Not more than one output should be shorted at a time.

5/ In $\overline{A}_i$ and $\overline{B}_i$, i = 0, 1, 2, 3.

6/ C_L = 50 pF ±10%, R_L = 2 kΩ ±5%.

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3.5 Electrical test requirements. Electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance, by device class, are specified in table II.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, the country of origin may be omitted from the body of the microcircuit but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*, 2, 3, 7, 9, 10, 11	1*, 2, 3, 7, 9	1, 7
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 9, 10, 11	1, 2, 3, 7, 9
Group B test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11	N/A	N/A
Group C end point electrical parameters (method 5005)	N/A	1, 2, 3	1
Additional electrical subgroups for group C periodic inspections	N/A	None	10, 11
Group D end point electrical parameters (method 5005)	1, 2, 3,	1, 2, 3	1

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification, and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D, or E using the circuit shown on figure 5, or equivalent.

2. T_A = 125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

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c. Percent defective allowable (PDA) - The PDA for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre-burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions, or equivalent:

1. Test condition D, or E, using the circuit shown on figure 4, or equivalent.

2. T_A = 125°C minimum.

3. Test duration: 1,000 hours except, as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

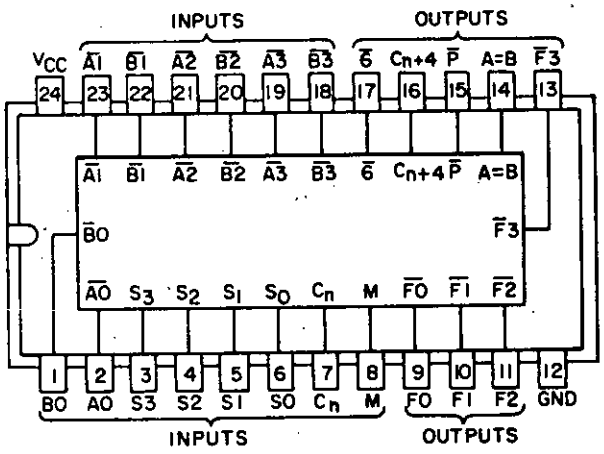
4.6 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510.

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Pin designations

Designation	Pin nos.	Function
$\overline{A}3, \overline{A}2, \overline{A}1, \overline{A}0$	19, 21, 23, 2	Word A inputs
$\overline{B}3, \overline{B}2, \overline{B}1, \overline{B}0$	18, 20, 22, 1	Word B inputs
S3, S2, S1, S0	3, 4, 5, 6	Function-select inputs
C_n	7	Inv. carry input
M	8	Mode control input
$\overline{F}3, \overline{F}2, \overline{F}1, \overline{F}0$	13, 11, 10, 9	Function outputs
A = B	14	Comparator output
$\overline{P}$	15	Carry propagate output
C_{n+4}	16	Inv. carry output
$\overline{G}$	17	Carry generate output
V_{CC}	24	Supply voltage
GND	12	Ground

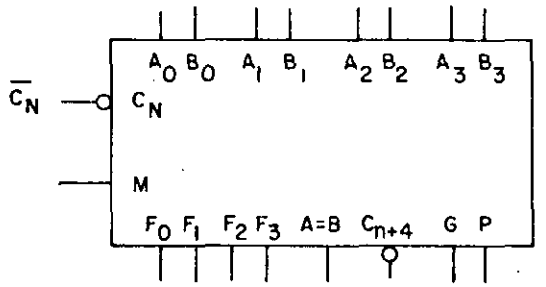
Cases J, K, L and Z



Device type 01

FIGURE 1. Terminal connections and pin designations.

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	Selection $S_3S_2S_1S_0$	Active high data 3/		
		M = H Logic functions	M = L; Arithmetic operations	
			C_n = 0 $\overline{C_n}$ = 1 = H	C_n = 1 $\overline{C_n}$ = 0 = L
0	L L L L	F = $\overline{A}$	F = A	F = A plus 1
1	L L L H	F = $\overline{A + B}$	F = A + B	F = (A + B) plus 1
2	L L H L	F = $\overline{A}B$	F = A + $\overline{B}$	F = (A + $\overline{B}$) plus 1
3	L L H H	F = 0	F = minus 1 (2's compl)	F = zero
4	L H L L	F = $\overline{AB}$	F = A plus A$\overline{B}$	F = A plus A$\overline{B}$ plus 1
5	L H L H	F = $\overline{B}$	F = (A + B) plus A$\overline{B}$	F = (A + B) plus A$\overline{B}$ plus 1
2/ 6	L H H L	F = A ⊕ B	F = A minus B minus 1	F = A minus B
7	L H H H	F = A$\overline{B}$	F = A$\overline{B}$ minus 1	F = A$\overline{B}$
8	H L L L	F = $\overline{A}$ + B	F = A plus AB	F = A plus AB plus 1
9	H L L H	F = $\overline{A \oplus B}$	F = A plus B	F = A plus B plus 1
10	H L H L	F = B	F = (A + $\overline{B}$) plus AB	F = (A + $\overline{B}$) plus AB plus 1
11	H L H H	F = AB	F = AB minus 1	F = AB
12	H H L L	F = 1	F = A plus A 1/	F = A plus A plus 1
13	H H L H	F = A + $\overline{B}$	F = (A + B) plus A	F = (A + B) plus A plus 1
14	H H H L	F = A + B	F = (A + $\overline{B}$) plus A	F = (A + $\overline{B}$) plus A plus 1
15	H H H H	F = A	F = A minus 1	F = A

1/ Each bit is shifted to the next more significant position.

2/ This device (ALU) can be used as a comparator when placed in the subtract mode (i.e., $S_3S_2S_1S_0$ are at logical levels LHHL respectively) and the following expressions are valid:

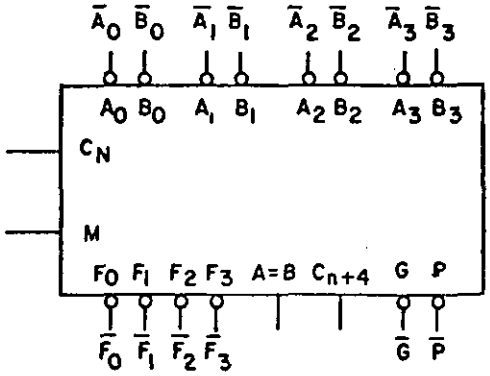
Active high data

When C_n is high and C_{n+4} is high, then A ≤ B
When C_n is low and C_{n+4} is high, then A < B
When C_n is high and C_{n+4} is low, then A > B
When C_n is low and C_{n+4} is low, then A ≥ B

3/ The table shown applies for positive logic. If negative logic is used, active high data becomes active low data.

FIGURE 2. Truth tables and logic equations for device type 01.

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	Selection $S_3S_2S_1S_0$	Active low data 3/		
		M = H Logic functions	M = L; Arithmetic operations	
			C_n = 0 = L	C_n = 1 = H
0	L L L L	$F = \overline{A}$	F = A minus 1	F = A
1	L L L H	$F = \overline{AB}$	F = AB minus 1	F = AB
2	L L H L	$F = \overline{A} + B$	F = $A\overline{B}$ minus 1	F = $A\overline{B}$
3	L L H H	F = 1	F = minus 1 (2's compl)	F = zero
4	L H L L	$F = \overline{A + B}$	F = A plus (A + $\overline{B}$)	F = A plus (A + $\overline{B}$) plus 1
5	L H L H	$F = \overline{B}$	F = AB plus (A + $\overline{B}$)	F = AB plus (A + $\overline{B}$) plus 1
2/ 6	L H H L	$F = \overline{A \oplus B}$	F = A minus B minus 1	F = A minus B
7	L H H H	$F = A + \overline{B}$	F = A + $\overline{B}$	F = (A + $\overline{B}$) plus 1
8	H L L L	$F = \overline{A}B$	F = A plus (A + B)	F = A plus (A + B) plus 1
9	H L L H	$F = A \oplus B$	F = A plus B	F = A plus B plus 1
10	H L H L	F = B	F = $A\overline{B}$ plus (A + B)	F = $A\overline{B}$ plus (A + B) plus 1
11	H L H H	F = A + B	F = A + B	F = (A + B) plus 1
12	H H L L	F = 0	F = A plus A 1/	F = A plus A plus 1
13	H H L H	F = $A\overline{B}$	F = AB plus A	F = AB plus A plus 1
14	H H H L	F = AB	F = $A\overline{B}$ plus A	F = $A\overline{B}$ plus A plus 1
15	H H H H	F = A	F = A	F = A plus 1

1/ Each bit is shifted to the next more significant position.

2/ This device (ALU) can be used as a comparator when placed in the subtract mode (i.e., $S_3S_2S_1S_0$ are at logical levels LHHL respectively) and the following expressions are valid:

Active low data

When C_n is low and C_{n+4} is low, then A $\leq$ B
When C_n is high and C_{n+4} is low, then A < B
When C_n is low and C_{n+4} is high, then A > B
When C_n is high and C_{n+4} is high, then A $\geq$ B

3/ The table shown applies for negative logic. If positive logic is used, active low data becomes active high data.

FIGURE 2. Truth tables and logic equations for device type 01 - Continued.

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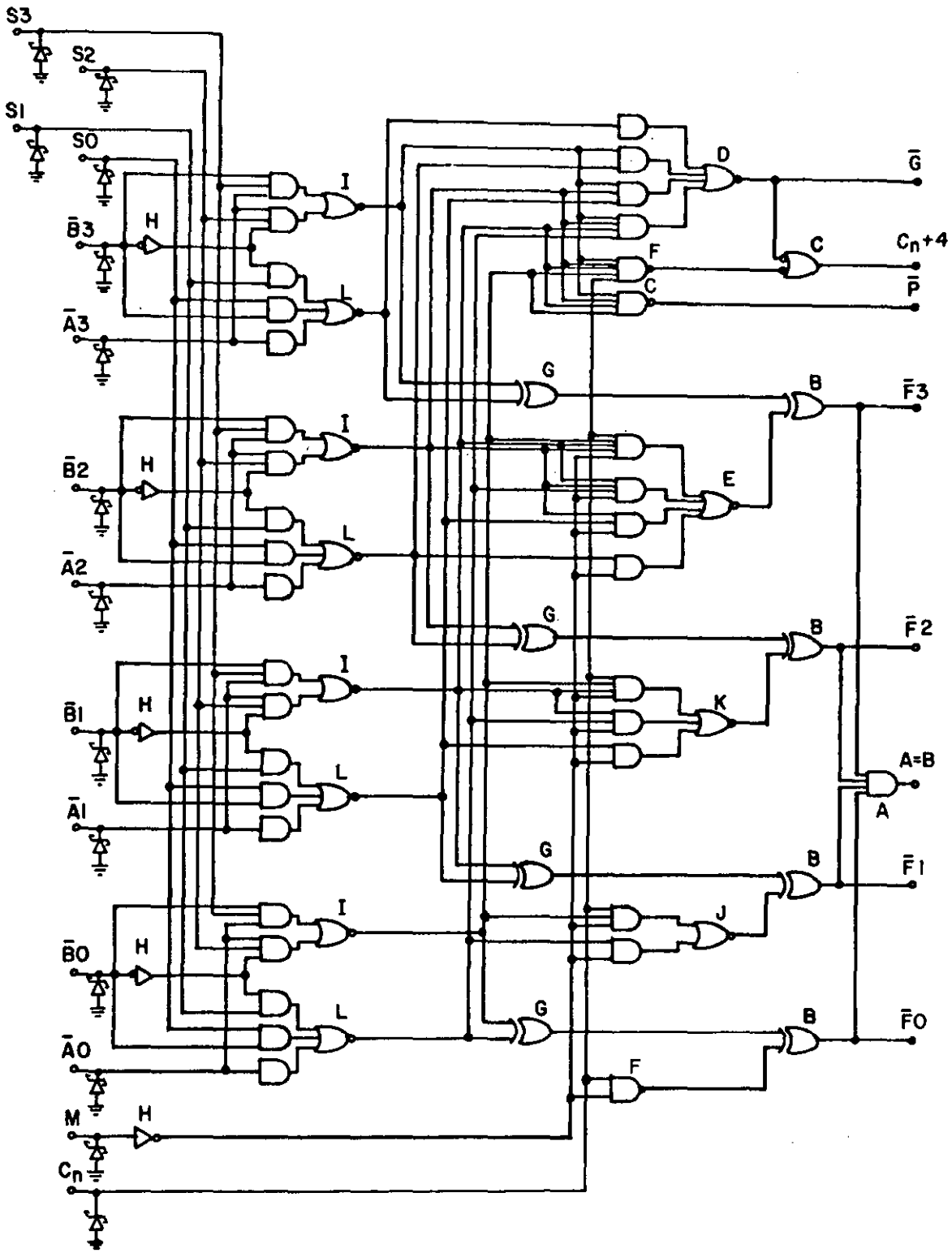
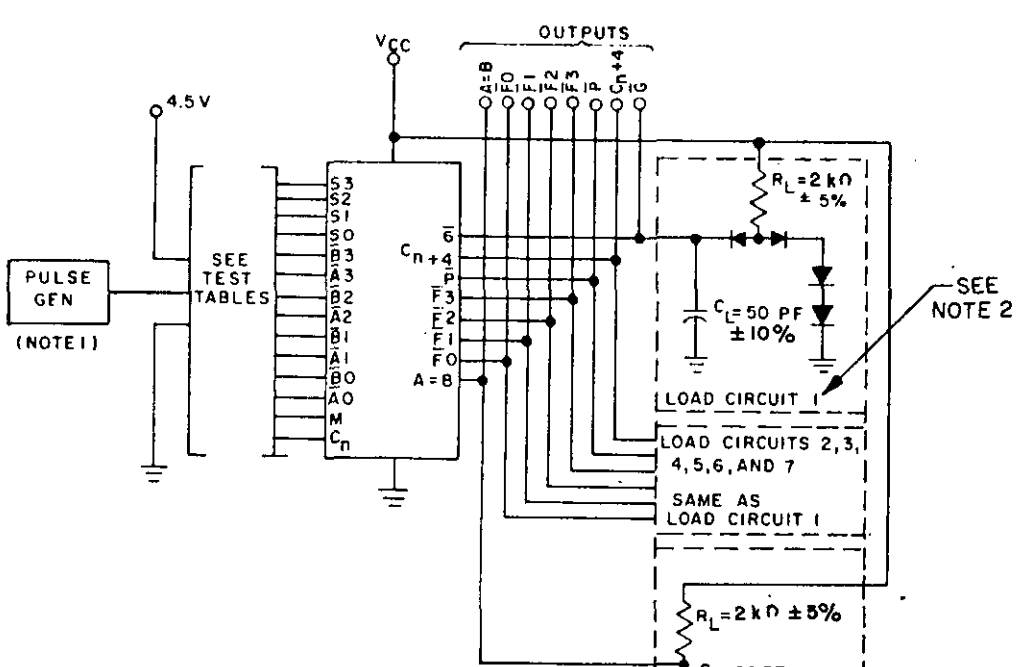
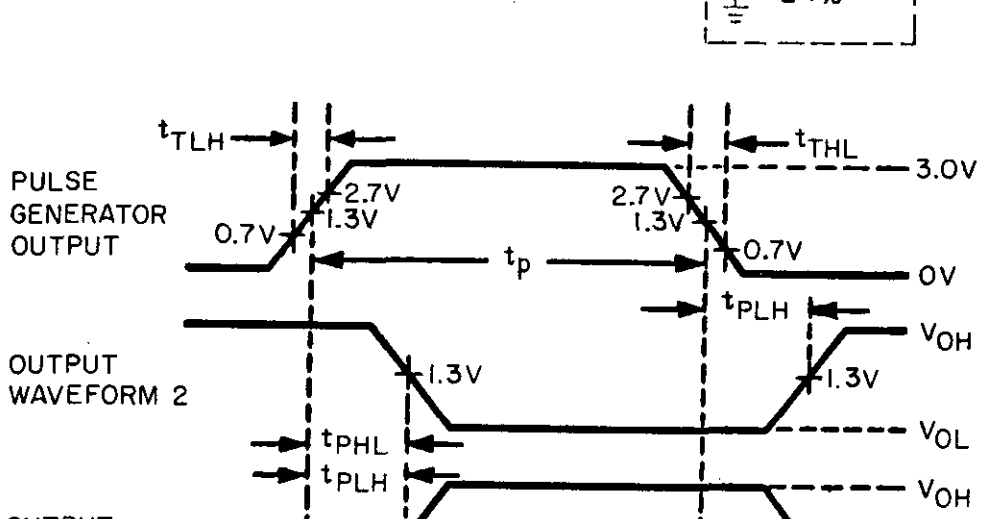


FIGURE 3. Logic diagram

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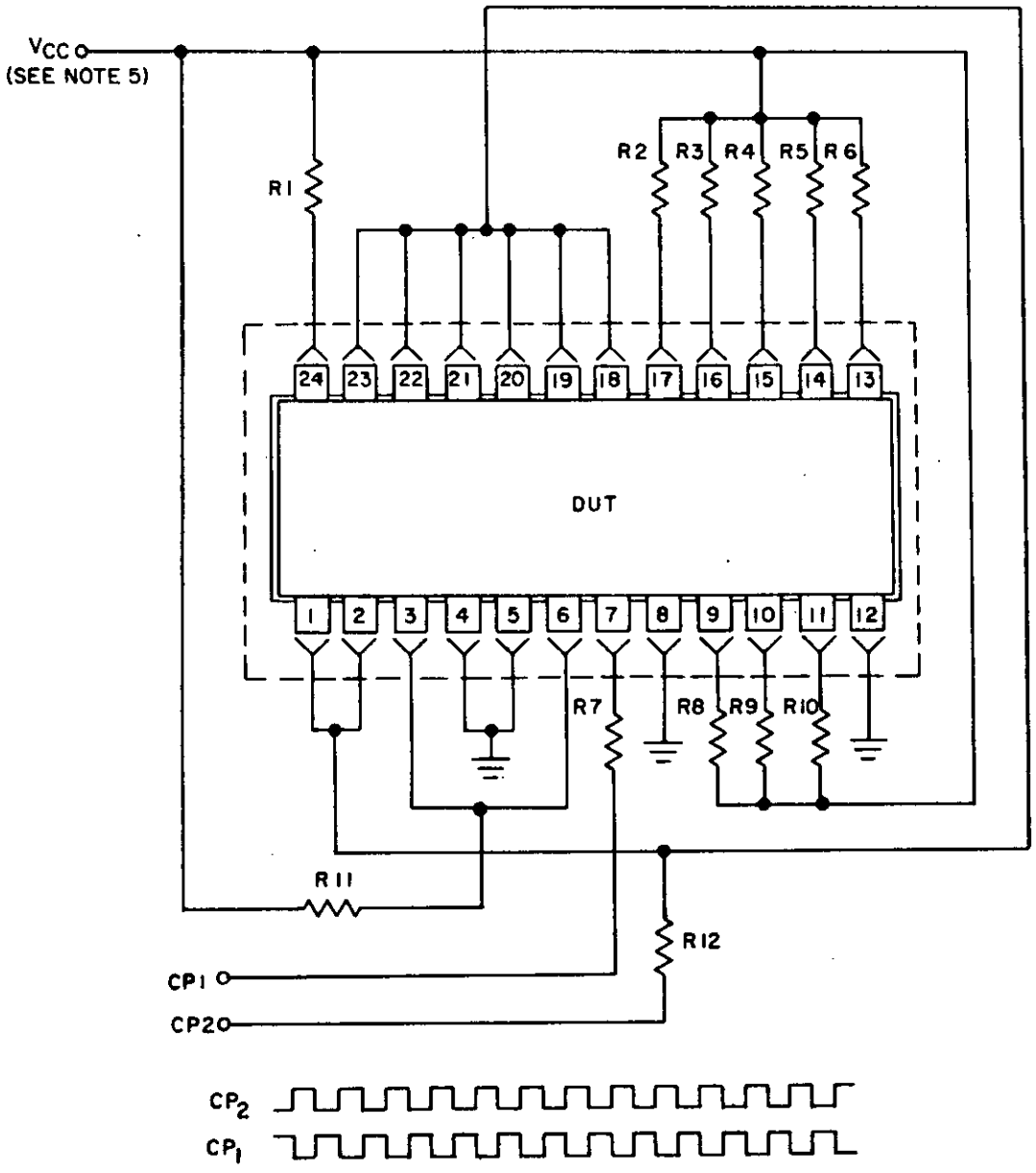




NOTES:
1. The pulse generator has the following characteristics: PRR = 1 MHz ±10%, $Z_{OUT} \approx 50$ ohms. Pulse width = 200 ns ±10%, $t_{TLH} \leq 15$ ns, $t_{THL} \leq 6$ ns.
2. Load circuits on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.
3. C_L includes probe and jig capacitance.
4. All diodes are 1N3064, or equivalent.

FIGURE 4. Waveforms for propagation delay time and test circuit for device type 01.

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NOTES:
1. R_1 = 5 ohms ±5%.
2. R_7 = 27 ohms ±5%.
3. R_2 thru R_6 = R_8 thru R_{11} = 1 kΩ ±5%.
4. R_{12} = 82 ohms ±5%.
5. V_{CC} shall be high enough to insure 5 V minimum at device terminals.
6. CP1 = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.
7. CP2 = Same as CP1 delayed by 1/2 cycle and synchronized with CP1.

FIGURE 5. Burn-in and life test circuit for device type 01.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be M ±2.0 V, or L ≤0.7 V, or open.

Subgroup	Symbol	MIL-STD-883 Method	Case L_1, L_2 Test no.	1 $\overline{B_0}$	2 $\overline{A_0}$	3 S_3	4 S_2	5 S_1	6 S_0	7 C_n	8 M	9 F_0	10 F_1	11 F_2	12 GND	13 F_3	14 A = B	15 $\overline{P}$	16 C_{n+4}	17 $\overline{G}$	18 B_3	19 A_3	20 B_2	21 A_2	22 B_1	23 A_1	24 V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
1 T_C = 25°C	V_{OH1}	3006	1	0.7 V	2.0 V	2.0 V	0.7 V	0.7 V	0.7 V	2.0 V					GND			-400 µA		-400 µA	0.7 V	2.0 V	0.7 V	2.0 V	0.7 V	2.0 V	4.5 V	G	2.5		V
	V_{OH2}		2	2.0 V			2.0 V		2.0 V												2.0 V		2.0 V		2.0 V			C_{n+4}			
	V_{OH3}		3					2.0 V																				A			
	V_{OH4}		4																									B			
	V_{OH4}		5																									F0			
	V_{OH4}		6																									F1			
	V_{OH4}		7																									F2			
	V_{OL1}	3007	8																									F3			
	V_{OL2}		9																												
	V_{OL3}		10																									G		0.4	
	V_{OL4}		11		0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	2.0 V	4 mA	4 mA	4 mA		4 mA		4 mA	4 mA	4 mA		0.7 V	0.7 V	0.7 V	0.7 V	0.7 V		C_{n+4}			
	V_{OL4}		12			2.0 V	2.0 V	2.0 V	2.0 V	2.0 V												2.0 V	2.0 V	2.0 V	2.0 V	2.0 V		A			
	V_{OL4}		13																									B			
	V_{OL4}		14																									F0			
	V_{OL5}		15																									F1			
	I_{IL1}	3009	16	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.4 V										5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	F2			
	I_{IL2}		17	0.4 V							5.5 V																	F3			
			18	5.5 V																								M	5/	5/	mA
			19							GND											0.4 V										
			20																												
	I_{IL3}		21						0.4 V	5.5 V											5.5 V	5.5 V						B0			
	I_{IL3}		22			0.4 V			5.5 V	5.5 V											5.5 V	5.5 V						B1			
	I_{IL4}		23	GND		GND		0.4 V	GND	GND	GND										GND	GND						B2			
	I_{IL4}		24				0.4 V	GND	GND	GND	GND										GND	GND						B3			
	I_{IL5}		25		GND		GND	GND	GND	0.4 V	GND										GND	GND	GND	GND	GND	GND		S0			
	I_{IL6}		26	5.5 V	0.4 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V										5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		S3			
			27	5.5 V	5.5 V																5.5 V	GND	5.5 V	0.4 V	5.5 V	0.4 V		S2			
			28	5.5 V	5.5 V																5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		S1			
			29	5.5 V	5.5 V																GND	0.4 V	GND	5.5 V	GND	5.5 V		S0			
	I_{IH1}	3010	30	GND	GND	GND	GND	GND	GND	GND	2.7 V										M	GND	GND	GND	GND	GND		M	20	µA
	I_{IH2}		31	2.7 V																								A0			
			32	GND						GND	GND												GND	GND	GND	GND			A1			
			33																				2.7 V	2.7 V	GND	GND			A2			
			34																				GND	GND					A3			
			35																										B0			
			36	2.7 V						5.5 V											2.7 V		2.7 V	GND					B1	60		
			37	GND																									B2			
			38																										B3			
	I_{IH3}		39			2.7 V			2.7 V	GND											GND	GND	GND	GND				S0			
	I_{IH3}		40						GND												GND	GND						S3			
	I_{IH4}		41	5.5 V		GND		2.7 V	GND												5.5 V	5.5 V	5.5 V	5.5 V				S1			
	I_{IH4}		42	5.5 V		GND	2.7 V	GND													5.5 V	5.5 V	5.5 V	5.5 V				S2	80		
	I_{IH5}		43	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	2.7 V	5.5 V										5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		C_n			
	I_{IH6}		44	GND	GND	GND	GND	GND	GND	GND	GND										GND	GND	GND	GND	GND	GND		M	100	100	
	I_{IH7}		45		5.5 V				5.5 V																			A0			
			46		GND				GND																			A1			
			47			5.5 V																						A2			
			48																									A3			
			49			GND	5.5 V	5.5 V	GND																			B0			
			50	5.5 V		GND			GND																			B1	300		
			51	GND																								B2			
			52																									B3			
	I_{IH8}		53			5.5 V			5.5 V																			S0			
	I_{IH8}		54						GND																			S3			
	I_{IH9}		55	5.5 V		GND		5.5 V	GND	GND											5.5 V	5.5 V						S1	400		
	I_{IH9}		56	5.5 V	5.5 V		5.5 V	GND	GND	GND											5.5 V	5.5 V						S2			
	I_{IH10}		57	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V										5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		C_n	500		

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, or L ≤0.7 V, or open.

Subgroup	Symbol	MIL-STD-883 method	Case J,K, L,2 Test no.	1 $\overline{B}_0$	2 $\overline{A}_0$	3 S_3	4 S_2	5 S_1	6 S_0	7 C_n	8 M	9 $\overline{F}_0$	10 $\overline{F}_1$	11 $\overline{F}_2$	12 GND	13 $\overline{F}_3$	14 A = B	15 $\overline{P}$	16 C_{n+4}	17 $\overline{G}$	18 $\overline{B}_3$	19 $\overline{A}_3$	20 $\overline{B}_2$	21 $\overline{A}_2$	22 $\overline{B}_1$	23 $\overline{A}_1$	24 V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
1 T_C = 25°C	I_{OS1}	3011	58 CKTS A, B and C	GND	GND	5.5 V	GND	GND	GND	GND	GND				GND					GND	5.5 V	5.5 V	GND	GND	GND	GND	5.5 V	G	-15	-100	mA
	I_{OS2}		59 CKTS A, B and C	GND	GND	GND	GND	GND	GND	GND	GND								GND		GND	GND	GND	GND	GND	GND		C_{n+4}			
	I_{OS3}		60 CKTS A, B and C 61 CKTS A, B and C 62 CKTS A, B and C 63 CKTS A, B and C 64 CKTS A, B and C	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	GND		GND		GND			5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V			$\overline{P}$ $\overline{F}_0$ $\overline{F}_1$ $\overline{F}_2$ $\overline{F}_3$			
	I_{CCL1}	3005	65	GND						GND											GND		GND		GND			V_{CC}		32	
	I_{CCH1}		66	GND	GND					GND											GND	GND	GND	GND	GND	GND		V_{CC}		35	
	V_{IC}		67 68 69 70 71 72 73 74 75 76 77 78 79 80	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA										-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	4.5 V	$\overline{B}_0$ $\overline{A}_0$ S_3 S_2 S_1 S_0 C_n M $\overline{B}_3$ $\overline{A}_3$ $\overline{B}_2$ $\overline{A}_2$ $\overline{B}_1$ $\overline{A}_1$		-1.5	V
	I_{CEX}		81		2.0 V	2.0 V	2.0 V	2.0 V	2.0 V		2.0 V						5.5 V					2.0 V		2.0 V		2.0 V		A=B		100	µA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C, and V_{IC} tests are omitted.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C, and V_{IC} tests are omitted.																														
7 1/2/4/ T_C = 25°C	Functional tests	3014	82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107	B B B A B B A B A B A

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open.

Subgroup	Symbol	MIL-STD-883 Method	Case outlines L, Z Test no.	1 B_0	2 A_0	3 S_3	4 S_2	5 S_1	6 S_0	7 C_n	8 M	9 F_0	10 F_1	11 F_2	12 GND	13 F_3	14 A = B	15 P	16 C_{n+4}	17 $\overline{E}$	18 B_3	19 A_3	20 B_2	21 A_2	22 B_1	23 A_1	24 V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
1/2/3/ T_C = 25°C	Functional tests	3014	108	B	B	A	B	B	B	A	A	H	H	H	GND	H	H	H	H	H	A	A	B	B	B	B	4.5 V	All outputs	H or L as shown 3/		
			109	B	A	B	B	B	A	A	B	H	H	H			H	L			B	B	A	A	A	A					
			110	A	B	A	B	A	B	B	A	H	H	L			L	H			A	A	B	B	B	B					
			111	B	A	A	A	A	B	A	B	H	L	H			H	L			B	B	A	A	A	A					
			112	A	B	B	A	A	A	B	B	H	L	L			L	H			A	A	B	B	B	B					
			113	B	A	A	A	B	A	B	A	H	L	H			H	L			B	B	A	A	A	A					
			114	A	B	B	A	A	B	A	B	H	H	L			L	H			A	A	B	B	B	B					
			115	B	A	A	A	B	B	A	A	H	H	H			H	L			B	B	A	A	A	A					
			116	A	B	B	A	A	A	B	A	L	L	H			L	H			A	A	B	B	B	B					
			117	B	A	A	A	B	B	A	B	L	L	L			H	L			B	B	A	A	A	A					
			118	A	B	B	A	A	A	B	B	L	H	H			L	H			A	A	B	B	B	B					
			119	B	A	A	A	B	B	A	A	L	H	L			H	L			B	B	A	A	A	A					
			120	A	B	B	A	A	A	B	A	L	H	H			L	H			A	A	B	B	B	B					
8	Same tests, terminal conditions and limits as for subgroup 7, except T_C = 125°C and -55°C																													
9 T_C = 25°C	t_{PHL1}	3003 Fig. 4	121	GND	IN GND	4.5 V	GND	GND	4.5 V	GND	GND	OUT	OUT	OUT	GND	OUT											15.0 V	F_0 to F_3 A_1 to A_3 B_1 to B_3	2	25	ns
			122																											
			123																											
			124																											
			125																											
			126																											
			127																											
			128																											
	t_{PLH1}		129		IN GND							OUT	OUT	OUT		OUT													55	
			130																											
			131																											
			132																											
			133																											
			134																											
			135																											
			136																											
	t_{PHL2}		137		IN 4.5 V	GND	4.5 V	4.5 V	GND			OUT	OUT	OUT		OUT													30	
			138																											
			139																											
			140																											
			141																											
			142		IN GND																									
			143																											
			144																											
	t_{PLH2}		145		IN 4.5 V							OUT	OUT	OUT		OUT													32	
			146																											
			147																											
			148																											
			149		IN GND																									
			150	GND																										
			151	GND																										
			152	GND																										
	t_{PHL3}		153	4.5 V	IN	4.5 V	GND	GND	4.5 V																					
			154	GND	GND																									
			155	GND	GND																									
			156	GND	IN																									
			157	GND	GND																									
			158	GND	GND																									
			159	GND	GND																									
			160	GND	GND																									
	t_{PLH3}		161	4.5 V	IN																								30	
			162	GND	GND																									
			163	GND	GND																									
			164	GND	IN																									
			165	IN	GND																									
			166	GND	GND																									
			167	GND	GND																									
			168	GND	GND																									

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, or L ≤0.7 V, or open.

Subgroup	Symbol	MIL-STD-883 method	Case U,K, L,Z Test no.	1 $\overline{B}_0$	2 $\overline{A}_0$	3 S_3	4 S_2	5 S_1	6 S_0	7 C_n	8 M	9 $\overline{F}_0$	10 $\overline{F}_1$	11 $\overline{F}_2$	12 GND	13 $\overline{F}_3$	14 A = B	15 P	16 C_{n+4}	17 G	18 $\overline{B}_3$	19 $\overline{A}_3$	20 $\overline{B}_2$	21 $\overline{A}_2$	22 $\overline{B}_1$	23 $\overline{A}_1$	24 V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
9 T_C = 25°C	t_{PHL4}	3003 Fig. 4	169 170 171 172 173 174 175 176	GND ↓ IN GND	IN GND GND GND 4.5 V GND GND GND	GND	4.5 V	4.5 V	GND	GND	GND				GND			OUT			GND ↓ IN	GND GND GND IN GND GND GND 4.5 V	GND ↓ IN GND	GND GND IN GND GND GND 4.5 V GND	GND ↓ IN GND	GND IN GND GND GND 4.5 V GND GND	5.0 V	$\overline{A}_0$ to P $\overline{A}_1$ to P $\overline{A}_2$ to P $\overline{A}_3$ to P $\overline{B}_0$ to P $\overline{B}_1$ to P $\overline{B}_2$ to P $\overline{B}_3$ to P	2	33	ns
	t_{PLH4}		177 178 179 180 181 182 183 184	↓ IN GND	IN GND GND GND 4.5 V GND GND GND																GND ↓ IN	GND GND GND IN GND GND GND 4.5 V	↓ IN GND	GND GND IN GND GND GND 4.5 V GND	↓ IN GND	GND IN GND GND GND 4.5 V GND GND		$\overline{A}_0$ to P $\overline{A}_1$ to P $\overline{A}_2$ to P $\overline{A}_3$ to P $\overline{B}_0$ to P $\overline{B}_1$ to P $\overline{B}_2$ to P $\overline{B}_3$ to P		30	
	t_{PHL5}		185 186 187 188 189 190 191 192	↓ 4.5 V 4.5 V 4.5 V IN 4.5 V 4.5 V 4.5 V	IN GND	4.5 V	GND	GND	4.5 V										OUT	4.5 V 4.5 V 4.5 V GND 4.5 V 4.5 V 4.5 V IN	GND GND GND IN GND ↓ IN GND	4.5 V 4.5 V GND 4.5 V 4.5 V 4.5 V IN 4.5 V	GND GND IN GND	4.5 V GND 4.5 V 4.5 V 4.5 V IN 4.5 V 4.5 V	GND IN GND		$\overline{A}_0$ to $\overline{A}_1$ to $\overline{A}_2$ to $\overline{A}_3$ to $\overline{B}_0$ to $\overline{B}_1$ to $\overline{B}_2$ to $\overline{B}_3$ to		28	
	t_{PLH5}		193 194 195 196 197 198 199 200	GND 4.5 V 4.5 V 4.5 V IN 4.5 V 4.5 V 4.5 V	IN GND																4.5 V 4.5 V 4.5 V GND 4.5 V 4.5 V 4.5 V IN	↓ IN GND	4.5 V 4.5 V GND 4.5 V 4.5 V 4.5 V IN 4.5 V	↓ IN GND	4.5 V GND 4.5 V 4.5 V 4.5 V IN 4.5 V 4.5 V	↓ IN GND		$\overline{A}_0$ to $\overline{A}_1$ to $\overline{A}_2$ to $\overline{A}_3$ to $\overline{B}_0$ to $\overline{B}_1$ to $\overline{B}_2$ to $\overline{B}_3$ to		32	
	t_{PHL6}		201 202 203 204 205 206 207 208	4.5 V GND GND GND IN GND GND GND	IN GND	GND	4.5 V	4.5 V	GND											GND GND GND 4.5 V GND GND GND IN	↓ IN GND	GND GND 4.5 V GND GND GND IN GND	↓ IN GND	GND 4.5 V GND GND GND IN GND GND	↓ IN GND		$\overline{A}_0$ to $\overline{A}_1$ to $\overline{A}_2$ to $\overline{A}_3$ to $\overline{B}_0$ to $\overline{B}_1$ to $\overline{B}_2$ to $\overline{B}_3$ to		30	
	t_{PLH6}		209 210 211 212 213 214 215 216	4.5 V GND GND GND IN GND ↓	IN GND															GND GND GND 4.5 V GND GND GND IN	↓ IN GND	GND GND 4.5 V GND GND GND IN GND	↓ IN GND	GND 4.5 V GND GND GND IN GND	↓ IN GND		$\overline{A}_0$ to $\overline{A}_1$ to $\overline{A}_2$ to $\overline{A}_3$ to $\overline{B}_0$ to $\overline{B}_1$ to $\overline{B}_2$ to $\overline{B}_3$ to		32	
	t_{PHL7}		217		4.5 V	4.5 V	4.5 V	GND	GND	4.5 V	IN		OUT							GND	4.5 V		4.5 V		4.5 V		$\overline{C}_n$ to $\overline{F}_0$		25	
	t_{PLH7}		218		4.5 V	4.5 V	GND	GND	4.5 V	IN		OUT												4.5 V		$\overline{C}_n$ to $\overline{F}_0$		26	
	t_{PHL8}		219 220 221 222 223 224 225 226	↓ IN GND GND GND	IN 4.5 V	GND	4.5 V	4.5 V	GND	GND							OUT			↓ IN	↓ IN 4.5 V	↓ IN GND	↓ IN 4.5 V	↓ IN GND	4.5 V IN 4.5 V		$\overline{A}_0$ to A=B $\overline{A}_1$ to A=B $\overline{A}_2$ to A=B $\overline{A}_3$ to A=B $\overline{B}_0$ to A=B $\overline{B}_1$ to A=B $\overline{B}_2$ to A=B $\overline{B}_3$ to A=B		75	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, or L ≤0.7 V, or open.

Subgroup	Symbol	MIL-STD-883 method	Case J,K, L,Z Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test Limits Min	Test Limits Max	Unit
				$\overline{B}_0$	$\overline{A}_0$	S_3	S_2	S_1	S_0	C_n	M	$\overline{F}_0$	$\overline{F}_1$	$\overline{F}_2$	GND	$\overline{F}_3$	A = B	P	C_{n+4}	G	$\overline{B}_3$	$\overline{A}_3$	$\overline{B}_2$	$\overline{A}_2$	$\overline{B}_1$	$\overline{A}_1$	V_{CC}				
9 T_C = 25°C	t_{PLH8}	3003 Fig. 4	227 228 229 230 231 232 233 234	GND ↓ IN GND GND GND	IN 4.5 V	GND	4.5 V	4.5 V	GND	GND	GND				GND		OUT				GND ↓ IN	4.5 V 4.5 V 4.5 V IN 4.5 V	GND ↓ IN GND	4.5 V 4.5 V IN 4.5 V	GND ↓ IN GND	4.5 V IN 4.5 V	5.0 V	$\overline{A}_0$ to A=B $\overline{A}_1$ to A=B $\overline{A}_2$ to A=B $\overline{A}_3$ to A=B $\overline{B}_0$ to A=B $\overline{B}_1$ to A=B $\overline{B}_2$ to A=B $\overline{B}_3$ to A=B	2	80	ns
	t_{PHL9}		235	GND		4.5 V	GND	GND	4.5 V	IN									OUT		GND		GND					C_n to C_{n+4}		25	
	t_{PLH9}		236	GND		4.5 V	GND	GND	4.5 V												GND		GND					C_n to C_{n+4}		26	
	t_{PHL10}		237	4.5 V		GND	4.5 V	4.5 V	GND												4.5 V		4.5 V		4.5 V			C_n to C_{n+4}		25	
	t_{PLH10}		238	4.5 V ↓		GND	4.5 V	4.5 V	GND																4.5 V			C_n to C_{n+4}		26	
	t_{PHL11}		239 240 241 242 243 244 245 246	GND 4.5 V 4.5 V 4.5 V IN 4.5 V 4.5 V 4.5 V	IN GND	4.5 V	GND	GND	4.5 V	GND											GND ↓ GND 4.5 V 4.5 V 4.5 V IN	GND GND GND IN GND	GND ↓ GND 4.5 V 4.5 V 4.5 V IN 4.5 V	GND GND IN GND	4.5 V GND 4.5 V 4.5 V IN 4.5 V 4.5 V	GND IN GND		$\overline{A}_0$ to C_{n+4} $\overline{A}_1$ to C_{n+4} $\overline{A}_2$ to C_{n+4} $\overline{A}_3$ to C_{n+4} $\overline{B}_0$ to C_{n+4} $\overline{B}_1$ to C_{n+4} $\overline{B}_2$ to C_{n+4} $\overline{B}_3$ to C_{n+4}		40	
	t_{PLH11}		247 248 249 250 251 252 253 254	GND 4.5 V 4.5 V 4.5 V IN 4.5 V 4.5 V 4.5 V	IN GND															4.5 V 4.5 V 4.5 V GND 4.5 V 4.5 V 4.5 V IN	GND ↓ IN GND	4.5 V 4.5 V GND 4.5 V 4.5 V 4.5 V IN 4.5 V	GND ↓ IN GND	4.5 V GND 4.5 V 4.5 V 4.5 V IN 4.5 V 4.5 V	IN GND		$\overline{A}_0$ to C_{n+4} $\overline{A}_1$ to C_{n+4} $\overline{A}_2$ to C_{n+4} $\overline{A}_3$ to C_{n+4} $\overline{B}_0$ to C_{n+4} $\overline{B}_1$ to C_{n+4} $\overline{B}_2$ to C_{n+4} $\overline{B}_3$ to C_{n+4}		38	
	t_{PHL12}		255 256 257 258 259 260 261 262	4.5 V GND GND GND IN GND GND GND	IN GND	GND	4.5 V	4.5 V	GND											GND GND GND 4.5 V GND GND GND IN	GND ↓ IN GND	GND GND 4.5 V GND GND GND IN GND	GND ↓ IN GND	GND 4.5 V GND GND GND IN GND GND	IN GND		$\overline{A}_0$ to C_{n+4} $\overline{A}_1$ to C_{n+4} $\overline{A}_2$ to C_{n+4} $\overline{A}_3$ to C_{n+4} $\overline{B}_0$ to C_{n+4} $\overline{B}_1$ to C_{n+4} $\overline{B}_2$ to C_{n+4} $\overline{B}_3$ to C_{n+4}		46	
	t_{PLH12}		263 264 265 266 267 268 269 270	4.5 V GND GND GND IN GND	IN GND															GND GND GND 4.5 V GND GND GND IN	GND ↓ IN GND	GND GND 4.5 V GND GND GND IN GND	GND ↓ IN GND	GND 4.5 V GND GND GND IN GND GND	IN GND		$\overline{A}_0$ to C_{n+4} $\overline{A}_1$ to C_{n+4} $\overline{A}_2$ to C_{n+4} $\overline{A}_3$ to C_{n+4} $\overline{B}_0$ to C_{n+4} $\overline{B}_1$ to C_{n+4} $\overline{B}_2$ to C_{n+4} $\overline{B}_3$ to C_{n+4}		40	
	t_{PHL13}		271 272 273 274 275 276 277 278	↓ IN GND GND	IN GND						4.5 V	OUT OUT	OUT OUT	OUT OUT		OUT OUT				GND ↓ IN	IN GND	↓ IN GND	IN GND	↓ IN GND	IN GND		$\overline{A}_0$ to $\overline{F}_0$ $\overline{A}_1$ to $\overline{F}_1$ $\overline{A}_2$ to $\overline{F}_2$ $\overline{A}_3$ to $\overline{F}_3$ $\overline{B}_0$ to $\overline{F}_0$ $\overline{B}_1$ to $\overline{F}_1$ $\overline{B}_2$ to $\overline{F}_2$ $\overline{B}_3$ to $\overline{F}_3$		38	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, or L ≤0.7 V, or open.

Subgroup	Symbol	MIL-STD-883 method	Case J,K, L,Z Test no.	1 B_0	2 A_0	3 S_3	4 S_2	5 S_1	6 S_0	7 C_n	8 M	9 F_0	10 F_1	11 F_2	12 GND	13 F_3	14 A = B	15 P	16 C_{n+4}	17 G	18 B_3	19 A_3	20 B_2	21 A_2	22 B_1	23 A_1	24 V_{CC}	Measured terminal	Test Limits Min	Test Limits Max	Unit
9 T_C = 25°C	t_{PLH13}	3003 Fig. 4	279	GND	IN	GND	4.5 V	4.5 V	GND	GND	4.5 V	OUT	OUT	OUT	GND						GND	GND	GND	GND	GND	GND	5.0 V	A_0 to F_0	12	33	ns
			280	↓	GND																GND	GND		GND		IN		A_1 to F_1			
			281																		GND	IN		IN		GND		A_2 to F_2			
			282									OUT				OUT					IN	GND		GND				A_3 to F_3			
			283	IN																	GND							B_0 to F_0			
			284	GND									OUT									IN	GND		IN			B_1 to F_1			
			285	GND								OUT		OUT								GND			GND			B_2 to F_2			
			286	GND												OUT					IN		IN					B_3 to F_3			
10 T_C = 125°C	Same tests and terminal conditions as subgroup 9, except T_C = 125°C and limits as follows:																													
	t_{PHL1} = 2 to 37 ns; t_{PLH1} = 2 to 55 ns;																													
	t_{PHL2} and t_{PLH2} = 2 to 55 ns;																													
	t_{PHL3} and t_{PLH3} = 2 to 52 ns;																													
	t_{PHL4} = 2 to 57 ns; t_{PLH4} = 2 to 52 ns;																													
	t_{PHL5} = 2 to 42 ns; t_{PLH5} = 2 to 51 ns;																													
	t_{PHL6} and t_{PLH6} = 2 to 55 ns;																													
	t_{PHL7} = 2 to 37 ns; t_{PLH7} = 2 to 46 ns;																													
	t_{PHL8} = 2 to 96 ns; t_{PLH8} = 2 to 100 ns;																													
	t_{PHL9} = 2 to 37 ns; t_{PLH9} = 2 to 48 ns;																													
	t_{PHL10} = 2 to 37 ns; t_{PLH10} = 2 to 48 ns;																													
	t_{PHL11} and t_{PLH11} = 2 to 64 ns;																													
	t_{PHL12} = 2 to 60 ns; t_{PLH12} = 2 to 69 ns;																													
	t_{PHL13} = 2 to 48 ns; t_{PLH13} = 2 to 57 ns.																													
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																													

1/ Tests shall be performed in sequence.

2/ Input voltages shown are: A = 2.0 volts minimum and B = 0.7 volts maximum.

3/ Output voltages shall be either: (a) H = 2.6 V, minimum and L = 0.4 V, maximum when using a high speed checker double comparator; or (b) H >1.5 V and L <1.5 V when using a high speed checker single comparator.

4/ All functional tests (82 thru 115) require a 1 kΩ pull up resistor between pin 14 and V_{CC}.

5/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Test no.	A	B	C	D
I_{IL1}	16	-0.16/-0.40	-0.03/-0.30	-0.16/-0.40	-0.15/-0.38
I_{IL2}	17,18,19,29	-0.48/-1.20	-0.09/-0.90	-0.48/-1.20	-0.45/-1.14
I_{IL3}	21, 22, 23, 24	-0.64/-1.60	-0.72/-1.44	-0.64/-1.60	-0.60/-1.52
I_{IL4}	21, 22, 23, 24	-0.64/-1.60	-0.72/-1.44	-0.64/-1.60	-0.60/-1.52
I_{IL5}	25	-0.8/-2.0	-1.0/-2.0	-0.8/-2.0	-0.75/-1.90
I_{IL6}	26, 27, 28, 29	-0.32/-0.80	-0.09/-0.09	-0.32/-0.82	-0.30/-0.76

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for original equipment design application and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirement for JAN marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Electrical ground (common terminal).

V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal.

V_{IC} - - - - - - - - - - - - - Input clamp voltage.

I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below in table IV is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed Generic-Industry type. Generic-Industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of Generic-Industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-Industry type
01	54LS181

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6.7 Manufacturers designation. Manufacturer circuits included in this specification are designated as shown in table IV herein.

TABLE IV. Substitutability and manufacturers designator.

Device type	A Texas Instruments	B Signetics Corporation	C Raytheon Company	D Motorola Incorporated
01	X	X	X	X

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-M-38510/308A Microcircuits, Digital, Low-Power Schottky TTL, Arithmetic Logic Units/Function Generators, Monolithic Silicon

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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